Supporting Information

Effect of sulfur on α-Al_2O_3-supported iron catalyst for Fischer-Tropsch synthesis

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Table S1. Peak deconvolution results of the H_2-TPR profile of Fe/ㅅ -Al_2O_3 catalyst.

Catalysts	Peak areas ($\times 10^6$)		
	Peak 1(< 700 K)	Peak 2 (700-940 K)	Peak 3 (> 940 K)
Fe/α-Al_2O_3	1.2	2.2	2.1

Table S2. Peak deconvolution results of the CO-TPR profiles of S_xFe/ㅅ -Al_2O_3 catalysts.

Catalysts	Peak areas ($\times 10^{-8}$)	
	Peak 1	Peak 2
Fe/α-Al_2O_3	1.2	14.6
$S_{0.001}$Fe/α-Al_2O_3	1.0	12.7
$S_{0.002}$Fe/α-Al_2O_3	1.1	15.0
$S_{0.005}$Fe/α-Al_2O_3	0.9	8.8
$S_{0.01}$Fe/α-Al_2O_3	1.0	6.0
$S_{0.02}$Fe/α-Al_2O_3	1.1	4.5
$S_{0.05}$Fe/α-Al_2O_3	1.2	4.0

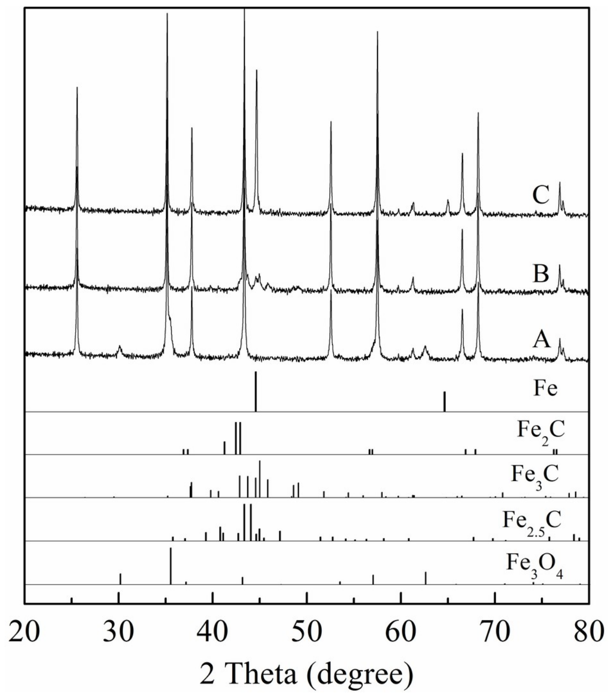

Fig. S1. XRD patterns recorded on the (A) Fe/α-Al_2O_3 catalyst after treating in a flow of CO/He = 5/95 mixture (v/v, 50 mL/min) at 573 K, (B) Fe/α-Al_2O_3 catalyst after treating in a flow of CO/He = 5/95 mixture (v/v, 50 mL/min) at 1073 K and (C) $S_{0.05}$Fe/α-Al_2O_3 catalyst after treating in a flow of CO/He = 5/95 mixture (v/v, 50 mL/min) at 1073 K.

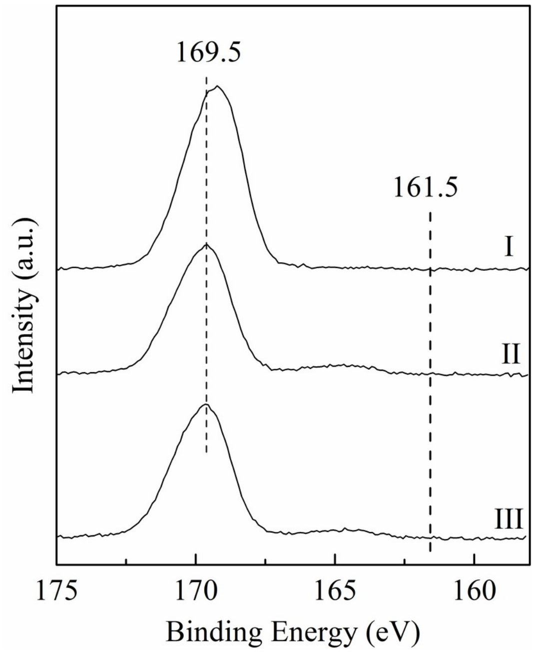

Fig. S2. S 2p XPS spectra of the 5wt% S/α-Al_2O_3 catalyst prepared by heating the sample (I) in a flow of O_2/N_2 = 21/79 mixture (v/v, 50 mL/min) at 673 K for 20 min, (II) in a flow of H_2 (50 mL/min) at 703 K for 120 min, (III) in a flow of H_2/CO = 1 mixture (v/v, 10 mL/min) at 573 K for 60 min.

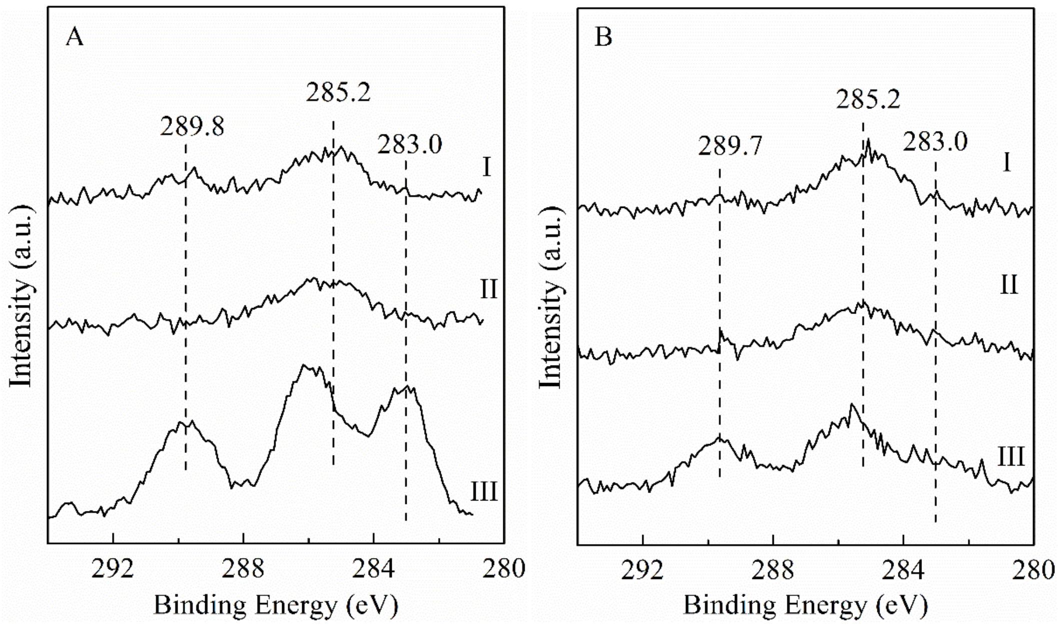

Fig. S3. C 1s XPS spectra of the (A) Fe/α-Al_2O_3 and (B) $S_{0.05}$Fe/α-Al_2O_3 catalysts prepared by heating the samples (I) in a flow of O_2/N_2 = 21/79 mixture (v/v, 50 mL/min) at 673 K for 20 min, (II) in a flow of H_2 (50 mL/min) at 703 K for 120 min,

(III) in a flow of H_2/CO = 1 mixture (v/v, 10 mL/min) at 573 K for 60 min.

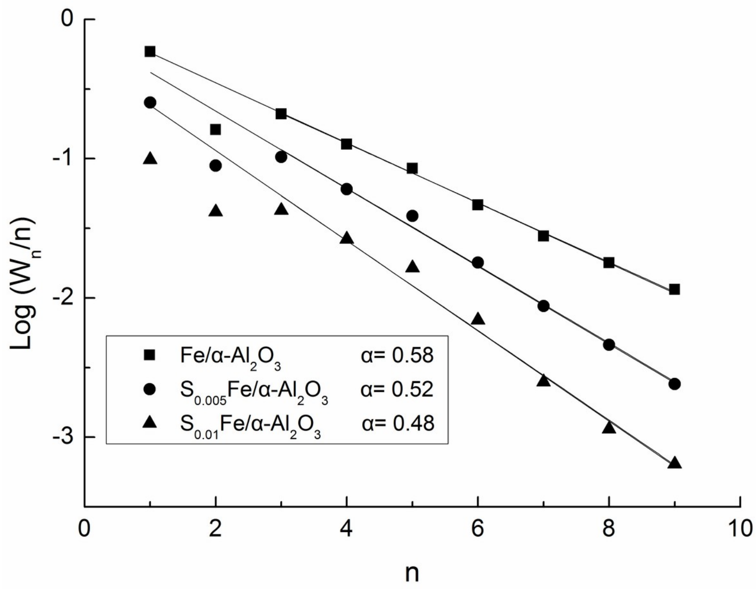

Fig. S4. Anderson-Schulz-Flory plots of the product distribution.

Reaction conditions: H_2/CO=1, 2.0 MPa, 24000 mL/(h · g_{cat}), 573 K, TOS = 7 h.